INCH-POUND

MIL-C- 83821 (84)

26 Nov 90

CRANE, WHEEL MOUNTED, CRASH, ROUGH TERRAIN, HYDRAULIC FULL REVOLVING, 4X4, DED, FIGHTER TYPE AIRCRAFT

This specification is approved for use within WR-ALC, Department of the Air Force,and is available for use by all Departments andAgencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel engine drive, wheel mounted, 4 X 4, four wheel drive, steering and brakes, 360° full revolving superstructure with cab, hydraulically operated crane. This crane will be used for on/off runway recovery of crash fighter type aircraft.

1.2 Classification. Cranes shall be of the following sizes as specified:

a. Size I - Crash recovery crane for up to and including F-15D size aircraft (see 3.2.2).
b. Size II - Crash recovery crane for up to and including F-15E and F-111F size aircraft (see 3.2.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131	Battery, Storage (Vehicular, Ignition, Lighting and Starting)
QQ-P-416	Plating, Cadmium (Electrodeposited)
VV-F-800	Fuel Oil, Diesel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMDETB, Robins AFB, GA 31098-5609, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3810

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-12459	Tire, Pneumatic: For Military Ground Vehicles
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, for Steel Surfaces
MIL-I-46058	Insulating Compound, Electrical (for Coating Printed Circuit Boards)
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-E-52649	Engine Cold Starting Aids, Ether Fuel Primers
MIL-S-81733	Sealing and Coating Compound, Corrosion Inhibitive
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-297	Rustproofing of Commercial (Nontactical) Vehicles
FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations for Use on Drawings, and on Specifications, Standards and Technical Documents
MIL-STD-130	Identification Markings of US Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-721	Definitions of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-808	Finishes, Materials and Processes for Corrosion Test Reports, Preparation of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Tests Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-965a	Parts Control Program
MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking, and Data Plate Standards
MIL-STD-1556b	Government/Industry Data Exchange Program (GIDEP) Contractor Participation Requirements
MIL-STD-1568	Material and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems
MIL-STD-1587	Material and Process Requirements for Air Force Weapon Systems

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

MIL-STD-1595	Qualification of Aircraft, Missile, and Aerospace Fusion Welders
MS 51117	Pintle Assembly, 10,000 Pounds Capacity, Manual Release

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US AIR FORCE

8838302	Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicle and Support Equipment

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DoD) adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be those in effect on the date specified in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INCORPORATED (ANSI)

B30.5	Mobile and Locomotive Cranes
Z26.1	Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Vehicles

(Application for copies should be addressed to American National Standards Institute, Incorporated, 1430 Broadway, New York NY 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B117	Salt Spray (Fog) Testing, Methods of

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103).

AMERICAN WELDING SOCIETY, INCORPORATED (AWS)

D1.1	Structural Welding Code, Steel

(Application for copies should be addressed to American Welding Society, Incorporated, 2501 NW Seventh Street, Miami FL 33125).

DEPARTMENT OF AGRICULTURE

5100-1A	Spark Arrestors for Internal Combustion Engines

(Application for copies should be addressed to Department of Agriculture Forest Service, Equipment Development Center, San Dimas CA 91773).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations (FMCSR) 393
Federal Motor Vehicle Safety Standard (FMVSS) 103

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington DC 20402).

POWER CRANE AND SHOVEL ASSOCIATION (PSCA)

PCSA Number 4	Mobile Power Crane and Excavator and Hydraulic Crane Standards

(Application for copies should be addressed to Construction Industry Manufacturer's Association, Marine Plaza, Suite 1700, 111 East Wisconsin Avenue, Milwaukee WI 53202).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J53	Minimum Performance Criteria for Emergency Steering of Wheeled Earth Moving Construction Machines
J88	Sound Measurement - Earthmoving Machinery - Exterior
J159	Load Moment Systems, Recommended Practice
J375	Radius of Load or Boom Angle Indicating Systems, Recommended Practice
J376	Load Indicating Device in Lifting Crane Service, Recommended Practice
J534	Lubrication Fittings
J537J	Storage Batteries
J765	Crane Load Stability Test Code
J820	Crane Hoist Line Speed and Power Test Code
J899	Operator's Seat Dimensions - Construction and Industrial Equipment Design
J919A	Sound Measurement - Earth Moving Machinery - Operator - Singular Type
J959	Lifting Crane, Wire Rope Strength Factors
J994	Alarm - Backup - Electric - Performance, Test and Application
J1063	Cantilevered Boom Crane Structures - Method of Test
J1129	Operator Cab Environment for Heated, Ventilated, and Air Conditioned Construction and Industrial Equipment
J1152	Minimum Performance Criteria for Braking Systems for off Highway, Rubber Tired, Self-Propelled Construction Machines
J1349	Engine Power Test Code, Spark Ignition and Diesel

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096).

TIRE AND RIM ASSOCIATION

Yearbook

(Application for copies should be addressed to Tire and Rim Association, Incorporated, 3200 West Market Street, Akron OH 44313).

NORTH ATLANTIC TREATY ORGANIZATION (NATO)

AEP-5

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.1.2 Materials. The selection of materials and processes shall be in accordance with MIL-STD-1568 and MIL-STD-1587. Materials not covered by this specification or referenced document shall be suitable for the intended environment and application. The contractor shall not use nonstandard or proprietary materials and processes without prior government approval.

3.1.2.1 Recycled materials. All materials, equipment and articles incorporated in the work covered by this specification shall be new.

3.1.2.2 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used. Surface preparation, finishes and coatings shall be in accordance with MIL-STD-808.

3.1.2.3 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.2.4 Primers. MIL-P-23377 and MIL-P-26915 epoxy primers shall be used on aluminium and ferrous alloys respectively.

3.1.2.5 Topcoat. MIL-C-83286 polyurethane shall be used as the topcoat finish.

3.1.2.6 Drainage. Complete drainage shall be provided for both internal and exterior pockets, areas, parts and components.

3.1.2.7 Magnesium. Magnesium and magnesium alloys shall be subject to approval by the government.

3.1.2.8 Prohibited Materials. The use of asbestos, wood and/or wood products is prohibited. Similarly, vinyl, polyvinylchloride, polyester, and corrosive type RTV (yields acetic acid upon curing) are prohibited for use in all electrical systems. If vinyl, polyvinylchloride, polyester, or corrosive type RTV are used near an electrical system, the area must be adequately ventilated so as to prevent any corrosive effects on the electrical wires or components.

3.1.2.9 Fasteners. Ferrous alloy fasteners shall be cadmium plated per QQ-P-416 Type II or zinc plated, to a minimum thickness of 0.0003 inches unless threaded tolerances dictate a different thickness; and aluminum alloy fasteners shall be anodized per MIL-A-8625. Fastener installation shall be in accordance with MIL-STD-1568. Fastener quality shall be in accordance with US Air Force Drawing 8838302.

3.1.2.10 Sealants for Permanent Fasteners. All joints, rivets and mating/faying surfaces of exterior and interior components shall be sealed using MIL-S-81733.

3.1.2.11 Conformal Coating. Printed circuit/printed wire boards shall be conformal coated with materials in accordance with MIL-I-46058.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.1.2.12 Welding. Ferrous and aluminum welding shall be IAW MIL-STD-1595.

3.1.2.13 Soldering. Soldering shall be IAW MIL-STD-454, requirement 5.

3.1.2.14 Brazing. Brazing shall be performed IAW MIL-B-7883.

3.1.2.15 Workmanship. Workmanship shall be IAW MIL-STD-454, requirement 9.

3.1.2.16 Commercial Off-the-Shelf Equipment. All commercial equipment and parts shall be assessed by the contractor as to their suitability to function in the intended environment and comply with the system's life expectancy requirements.

3.1.2.17 Galvanized steel. Many producers of zinc-coated steel apply inhibitors to protect their products from moisture during shipment and storage. Some of these inhibitors interfere with pretreatments resulting in poor priming surfaces. Contractors shall consult with their supplier of galvanized steel for recommendations on the pretreatment to be used on their respective product.

3.1.3 Design and construction. The crane shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the crane shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, removal and replacement. The crane shall comply with all requirements of 29CFR 1926.550, SAE J765, SAE J1063 and ANSI B30.5.

a. Finishes. Protective coating application shall be considered during the design stage. Sharp, angular corners are undesirable due to coating failures on outside corners (too thin) and the tendency of coatings to bridge inside corners. Smooth, round corners allow uniform coating thickness and avoids creation of possible sump areas. All mechanically joined components shall be finished before assembly. Welded structures shall have finishes applied after welding. When welded and bolted assemblies are used, manufacturing steps shall be arranged to minimize disruption of the protective coatings. Supplementary corrosion prevention treatments - rustproofing, sealants, greases, etc., shall be applied after finish application. All metallic surfaces shall be treated for corrosion prevention.

b. Joints and Faying surfaces. Joint designs shall minimize crevices formed by overlapping surfaces and shall point away from the forward direction of vehicle travel or point down. Butt weld are preferable to lap welds. Continuous welds are preferable to tack welds. The joints shall be designed so there are no gaps on joint laps. This can be accomplished by continuous welds, solder or sealant bead lines. All faying surfaces shall be primed and completely coated with sealant prior to assembly.

c. Core panel structures. Pretreat and prime all internal surfaces prior to assembly. Pretreat and prime interior panel skin and faying surfaces prior to assembly. Coat entire faying surface area with sealant during assembly and apply sealant bead after assembly to the lap joint. The panels shall then be foamed in placed using non-reverting polyurethane.

d. Drain holes. Structures shall be designed to avoid pockets where debris or moisture might accumulate. Inverted u-channel structural members do not accumulate moisture or debris. Closed structural configurations should have sloping surfaces towards drain holes which are adequately sized and placed in the lowest sump areas. All interior surfaces shall be treated IAW MIL-STD-808, with priming being the minimum finish requirement. The finish shall be reapplied/applied through drain holes after welding operations.

e. Doors. Consideration shall be given to open configuration doors. Use of single panel doors eliminate crevice corrosion sites and leads to easier maintenance of latches and window mechanism.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

f. Alternate materials. Use of composites, polymers, etc., type materials should be investigated. While not subject to corrosion, consideration of material interaction, environmental stability and field repair should be investigated before recommending alternate material.
g. Shields. Wheeled vehicles shall have aprons or splash shields designed to prevent mud, stones, debris and water from being thrown against areas outside of the wheel-wells.

3.1.3.1 Reliability. The vehicle including all components, shall have a minimum Mean-Time-Between-Failure (MTBF) of not less than 250 hours of operation demonstrated at a 0.80 confidence level in accordance with 4.6.7.2 and 4.6.7.3. A failure is defined as the event or inoperable state in which any item, or part of an item, does not or would not perform as previously specified. Definitions of reliability terms not included herein shall be as found in MIL-STD-721 and MIL-STD-781. In the event of a conflict between these documents, MIL-STD-781 shall take precedence. The vehicle shall be designed for a life expectancy of at least 15 years.

3.1.3.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.
b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.
c. Ease of adjusting, servicing and replacing parts and components.
d. Use of readily available standards tools and equipment for maintenance.
e. Maintenance with a minimum number of tools.
f. Mean corrective maintenance downtime of no greater than one hour.
g. Maximum corrective maintenance downtime of no greater than four hours.
h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 35 minutes.

3.1.3.2.1 Maintenance tasks. The vehicle shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

a. Remove, replace and adjust all engine driven belts - 0.75 hour.
b. Remove and replace alternator - 0.5 hour.
c. Remove and replace all filters, screens and strainers in all hydraulic systems - 1 hour.
d. Remove and replace engine coolant system hoses - 0.5 hour.
e. Drain engine lubricating oil, remove and replace oil filter element(s) and refill crankcase - 1 hour.
f. Remove and replace fuel filter element(s) - 0.5 hour.
g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 0.5 hour.
h. Drain torque converter oil and transmission oil, remove and replace air filter elements and strainers and refill converter and transmission - 1 hour.
i. Remove and replace starter - two mechanics - 2.5 hours.

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss or torque or adjustment. Cotter pins, lock washers or nylon patches shall not be used for this purpose, except for attachment of trim items or as provided in commercial (see 6.3.1) components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.1.3.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility. The crane shall be designed and constructed to operate satisfactorily when lubricated with the standard military lubricants specified in MIL-M-8090, except in Table II where hydraulic system shall be manufacturer's standard commercial hydraulic oil.

3.1.3.7 Component protection. All space in which work is performed during operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.3.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.4 Environmental conditions. The crane, including electrical switches, hydraulic cylinders, pumps, winches, etc, shall be capable of satisfactory storage and operation under the following environmental conditions:

a. Full-time exposure to temperatures ranging from -65°F to +125°F.
b. Full-time exposure to relative humidities up to 100 percent.
c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 mm to 4.5 mm falling at a rate of 5.2 inches per hour.
d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80cm^2/hour.
e. Part-time (five percent of the time) exposure to blowing dust; velocity 1750 ft/min, concentration 0.5 g/ft^3.
f. Part-time (five percent of the time) exposure to blowing sand; velocity 5700 ft/min, concentration 0.0773 g/ft^3.

3.1.5 Weight distribution. With lubricating and cooling systems full, fuel tank(s) at least half full with crane resting on firm, level and a uniform supporting surface, with outriggers unsupported, with the boom fully retracted and in travel position; the weight on the front axle shall be within 40 percent to 60 percent of the total weight of the crane and shall conform to the following additional requirements:

a. With the longitudinal axis of the rotating superstructure of the crane at 90° to the longitudinal axis of the chassis, the boom retracted and set at the minimum recommended radius, the total load on all wheels and on the side of the chassis under the boom shall not be less than 15 percent of the total weight of the crane.
b. With the longitudinal axis of the rotating superstructure of the crane in line with the longitudinal axis of the carrier, in either direction, the boom retracted and set at the minimum recommended radius, the total load on all wheels under the lighter loaded end of the chassis shall not be less than 15 percent of the total weight of the crane.

3.1.6 Electromagnetic interference. The vehicle shall comply with the requirements of MIL-STD-461, Class C1, Group II for electromagnetic interference, emissions, and susceptibility, except that the limits for UM03 (Figure 8-1) shall be relaxed by 20dB.

3.1.7 Sound levels. The sound level, at the operator's ear level, shall not exceed 84dB (during normal range of operation) in the cab when measured in accordance with SAE J919a and outside the cab when measured in accordance with SAE J88.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.1.8 Boom. The boom shall be of the hydraulically operated telescoping type, consisting of a case section and sliding sections, capable of extension and retraction to the specified lengths. Elevation of the boom shall be accomplished by use of hydraulic cylinders. Telescoping and elevating of the boom do not have to occur simultaneously. The boom tail shall not project more than 23 inches beyond the front of the carrier or more than nine inches beyond the rear of the carrier in any operating condition. A boom angle indicator of the electronic type shall be provided. Upon hydraulic failure, the boom shall maintain boom angle and not permit uncontrolled lowering of the boom in accordance with ANSI B30.5. Boom sections shall be marked with various colors every three feet (bottom and cab side) for the entire length to ensure equal extension of boom sections. Minimum extended boom length of Size I and Size II cranes shall be 80 feet.

3.2 Performance.

3.2.1 Mobility. The fully loaded and equipped vehicle shall:

a. Travel speed. The unloaded crane shall be capable of a forward speed of at least 16 miles per hour (MPH) and a reverse speed of at least 4 MPH.
b. Pick and carry speed. The crane shall be capable of carrying the rated pick and carry load at a forward or reverse speed of at least 2.5 MPH.
c. Cross country operation. The unloaded crane shall be capable of operation over cross country terrain, the area adjacent to aircraft runways taxiways, and parking aprons with a slope of no greater than 10 percent, where negotiation of small drainage ditches, timbers, rocks, etc. shall not be necessary, at an average speed not less than 2.5 MPH.
d. Turning radius. The turning radius for the Size I and Size II crane shall be no greater than 285 inches.
e. Gradeability. The cranes shall climb a 30 percent (minimum) grade on paved or hard packed surfaces and negotiate 15 percent side slopes.
f. Terrain. Cranes shall be able to extend outriggers and lift on a 6 percent slope (side or front).
g. Boom rotation speed. Boom shall independently and simultaneously lift and travel with a sweep of 360° continuous rotation at a maximum rotational speed of not less than 1.25 revolutions per minute (RPM). Swing speed within this range shall be smooth and constant, and adequate control shall be provided in the hydraulic circuits to insure smooth starting and stopping. The crane shall be capable of simultaneously lifting and rotating, throughout 360°, a load equal in pounds to the line pull determined from the condition as indicated herein. The swing mechanism shall be equipped with a swing brake (parking and dynamic) and swing lock in accordance with PSCA Number 4.
h. Load control. Means for positive control of any rated load; lowering or lifting at any boom length or elevation shall be provided. Hoist winch shall raise, lower or hold not less than permissible line pull without slippage on a single part line.
k. Negotiate eight inch high blocks with diagonally opposite wheels of the vehicle.

3.2.2 Rated capacity. The crane shall have a rated capacity as specified at the designated radii. The capacity on outriggers and on rubber shall be for 360° rotation (except where noted). Pick and carry capacity shall be with the boom centered over the front and shall be at least equal to the specified capacities on rubber. The crane shall comply with the requirements of SAE J765 for stability and SAE J1063 for structural integrity at the specified loads. The Size I crane shall not have outrigger interference at 8 foot elevation of load during a 360° rotation (without boom angle change) of a F-4 and F-15 weighing up to 45,000 pounds, and a Size II crane shall not have outrigger interference at 8 foot elevation of load during a 360° rotation of an F-111 (wings extended) or F-15E weighing up to 60,000 pounds. The weight of the hook and block shall be in addition to the rated capacities listed below:

	SIZE I		SIZE II	
WORK RADIUS IN FEET	360° WITH OUTRIGGERS	360° ON RUBBER	OVER THE FRONT WITH OUTRIGGERS	360° ON RUBBER
12	106,500	39,890	119,600	67,825
15	103,000	38,000	114,730	49,300

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:47Z
Check the source to verify that this is the currentversion before use.

20	80,000	23,350	99,410	30,620
25	62,000	15,600	80,000	20,885
30	45,000	10,825	77,600	15,065
31			77,600	
35	39,000	7,625	52,900	10,890

When lifting aircraft (at maximum weights stated in the paragraph), over the front and with outriggers, both types of cranes shall maintain at least three feet clearance between aircraft and crane. Both cranes shall be able to lift aircraft (at maximum weights stated in 3.3.2) to a minimum height of nine feet measured from bottom of fuselage to ground.

3.3 Carrier.

3.3.1 Powertrain.

3.3.1.1 Engine. The vehicle shall be equipped with a commercial (see 6.3.1) liquid cooled diesel engine of not less than six cylinders. The total power output required by all items driven by the engine shall not exceed its net power output rating at 2,000 feet elevation, determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when operating on diesel fuel in accordance with VV-F-800 and Grade JP-8 turbine fuel, the primary fuel, in accordance with MIL-T-83133. No change in any engine component is allowed. The engine shall be certified that it has successfully completed NATO AEP-5 test procedures using only JP-8 fuel. Any engine adversely affected by any one of the above fuels is not acceptable. The engine manufacturer's published ratings for the engine shall not be arbitrarily raised to meet the requirements of this specification. Copy of engine manufacturer's certification to meet fuel requirements shall be provided with the end item. The engine torque, speed, cubic inch displacement, and horsepower curve shall be compiled and prepared by the engine manufacture and be submitted with the end item.

3.3.1.1.1 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein; coolant temperature shall not exceed the engine manufacturer's recommended maximum. Expansion tanks of sufficient capacity for expansion of all coolant in the system shall be provided.

3.3.1.1.2 Engine starting system. The engine shall start within 15 seconds, after use of starting aids when required, in any ambient temperature within the required operating range of the vehicle. Installed glow plugs, fluid starting aids and heat from the winterization system may be used to facilitate engine starting under the following conditions:

TEMPERATURE RANGE (DEGREES F)	STARTING AIDS PERMITTED
40 through 125	None
1 through 39	Glow plug and fluid starting aids
-40 through 0	Glow plugs, fluid starting aids and heat from the winterization system

The starting system shall be equipped with an interlock device to prevent starter engagement when the transmission is in any position other than neutral.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.3.1.1.2.1 Ether injection. An ether injector, in accordance with MIL-E-52649A, Type III, shall be provided. It shall be of the measured shot type and shall have thermal controls which protect the engine against ether injection after warmup.

3.3.1.1.3 Exhaust system. The exhaust system shall be designed to expel exhaust gases in a manner that is not hazardous to operators or mechanics; the outlet shall not be directed toward the ground. The exhaust system, including all fasteners, shall be constructed of corrosion resistant materials. The system shall be protected from damage which could result from traversing rough terrain. Spark arrestors, meeting Department of Agriculture 5100-1A, shall be included in the exhaust system.

3.3.1.2 Transmission. The crane shall be equipped with a torque converter and a power shift or fully automatic transmission providing not less than three forward and two reverse gear ratios. The transmission shall be compatible with the engine and driveline components. The transmission shall have a maximum input rating greater than the maximum net torque and power output of the engine. Transmission shall have three speeds in low range (4 X 4 drive). The transmission shall provide for shifting under full engine power through all successive gear ratios in each direction. Transmission shall provide controlled applications of the clutches to provide smooth shifting without slippage that could damage the clutches. Oil pressure for actuation of the transmission clutches, for torque converter supply and for transmission lubrication, shall be supplied by a pump. Transmission and torque converter fluid shall be filtered by full-flow filter(s) having replacement elements. Selective forward and reverse directional controls activated by the operators foot are not acceptable. Transmission controls shall be accessible to the operator wearing arctic or chemical warfare clothing. Transmission controls shall be detented or controlled gear selection. Egress and ingress shall be accomplished without hitting the transmission or any other controls.

3.3.1.3 Axles. The front and rear axles shall be of sufficient capacity to sustain the imposed loads and torques imposed by all operations specified herein. Maximum practical component interchangeability shall be maintained between the front and rear axles. The rear axle shall be of the oscillating type.

3.3.2 Steering. The crane shall be equipped with an 18 inch automotive type steering wheel to actuate a hydraulic power assisted or fully hydraulic steering system. The steering system shall provide two wheel steer, four wheel steer, and oblique steer modes. A steering system indicator light shall be installed in the dash advising the operator of the crane's mode of steer. A separate indicator light shall be installed to let the operator know when the rear wheels are centered. All steering systems shall not require more than six turns to move the steering wheels from one stop to the opposite stop. All steering shall require not more than eight seconds for the maximum range of movement at low idle and not more than four seconds at governed engine speed. A minimum of four inches shall be between the steering wheel and any control surface. Provisions shall be made for adjustment to compensate for component wear. Lubrication fittings shall be provided at all friction points. Steering for the front wheels shall be designed to maintain manual steering in the event of engine or hydraulic failure. Steering system shall provide for retaining steering control at any speed and shall comply with SAE J53.

3.3.3 Brakes.

3.3.3.1 Service brakes. Air actuated disc or drum type service brakes shall be provided. The service brakes shall hold the vehicle on a 25 percent slope per SAE J1152. The vehicle shall be capable of stopping per SAE J1152 from its maximum speed on a clean, level and paved surface. The service brakes shall comply with SAE J1152 and ANSI B30.5. The air brake system shall have an air dryer and conform to DOT FMCSR, Sections 393.40 through 393.42, 393.43, and 393.45 through 393.52.

3.3.3.2 Parking brakes. Mechanical or spring set parking brake shall be provided. The parking brakes shall comply with and perform with SAE J1152. The parking brakes shall be capable of holding the crane with the pick and carry load on a longitudinal slope of not less than 30 percent without slippage.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

3.3.4 Frame. The frame assembly shall provide a rigid unit structure and shall be of bolted, riveted and/or welded steel construction. The frame shall provide stable retention for the boom in all positions and shall withstand all forces resulting from lifting, swinging and traveling with the charted safe loads.

3.3.5 Suspension system.

3.3.6 Wheels and tires. The crane shall be equipped with commercial high flotation tires and wheels which meet all Tire and Rim Association requirements for this application. If tube type tires are provided, inner tubes shall comply with Tire and Rim Association requirements. When required (see 6.2), a spare wheel and tire shall be provided; also, tires supplied shall conform to MIL-T-12459 for ozone resistance and identification when required (see 6.2).

3.3.7 Fuel system.

3.3.7.1 Fuel tank. A fuel tank shall be provided with a capacity capable of allowing at least 10 continuous operating hours at governed speed.The tank shall have a fill opening of three inches minimum, and readily accessible to personnel standing on the ground. The fuel tank shall be located and mounted so as to provide maximum protection from damage. A drain pet cock shall be provided at the lowest point of the fuel tank. Drain cock shall be guarded against inadvertent or accidental damage.

3.3.7.2 Fuel filter. At least one primary filter/strainer (with water drain valve) of the fuel water separator type shall be located in the fuel line between the fuel tank and the fuel pump. At least one secondary filter for filtering fine dirt particles shall be located on the pressure (outlet) side of the fuel pump. All fuel filters shall be installed so that the elements can be readily serviced without disturbing other items. If located below the fuel tank level, a shutoff valve shall be provided between it and the tank so that the element may be changed without draining the tank.

3.3.8 Hydraulic system. The hydraulic system shall include the pump, cylinders, reservoir, filter(s), pressure relief valves, and all necessary lines, restrictors, and control valves to ensure positive control of all operational functions. In the event of total engine or hydraulic failure, a fail safe system shall not allow the system to drop more than 0.25 inch or allow rotation of more than 0.5 degree. A means shall be furnished to manually bypass the fail safe system to safely lower the boom or return the vehicle to travel mode. The pump shall be driven by the crane engine by positive gear drive. The pump shall have sufficient capacity to energize all hydraulically powered components, except steering and outriggers, operating simultaneously. Provisions shall be made for bleeding all air from the system. All hydraulic control valves and relief valves, when possible, shall be banked together for ease of maintenance. All hydraulic hoses and fittings shall be capable of withstanding a bursting pressure of four times the working pressure and a proof pressure of two times the maximum operating pressure.

3.3.8.1 Hydraulic reservoir. Hydraulic reservoir shall include a filler neck at least 2.25 inches with a filter screen and captive cap attached in or on the hydraulic tank neck. Reservoir shall have sufficient capacity to prevent air from entering the system with all hydraulic cylinders extended and sufficient free air capacity to prevent oil from being discharged through the air vent. A sight gauge shall be installed on the reservoir for fluid level readings. The location of the reservoir shall not limit access to any other component. The reservoir shall have one or more access holes with covers for inspection, cleaning, and maintenance. Hydraulic reservoir shall incorporate a manual means of preventing fluid loss during maintenance of hydraulic system. Hydraulic system shall contain an oil cooler (heat exchanger) which maintains the hydraulic oil temperature below the manufacturer's recommended temperature at all times.

3.3.8.2 Hydraulic filtration. System filtration shall be accomplished by a 10 micron filter located on the pressure side of the pump.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

3.3.8.3 Hydraulic controls. Controls shall be self-centering for all hydraulic functions. Force necessary to move any controls shall not exceed 15 pounds at the hand grips. Total travel of control levers shall not exceed 10 inches. Clearance of all around control lever hand grips shall be not less than 2.5 inches in any control position to accommodate arctic and chemical warfare gloves. Control levers shall not interfere with egress/ingress of the operator.

3.4 Cab. A fully enclosed cab shall be provided with ROPS (Roll-Over Protection System) and FOPS (Falling Objects Protection System) protection. The cab shall be a rigid, watertight compartment. The cab shall be of sheet steel construction and shall have tinted safety glass on all four sides and in the roof for maximum visibility. All glass shall be at least 0.25 inch nominal thickness, safety type, conforming to ANSI Z26.1, Type ASI. At least one window on each side and one rear window shall be capable of opening and being held in the open position while operating, for ventilation. The cab door latch shall be sufficient to prevent the door from accidentally opening and shall be operable by a person wearing heavy arctic mittens. A nonskid surface (e.g., diamond plate) on the cab floor located under the brake and accelerator pedals shall be furnished. The cab shall provide access holes in its floors for any adjustment on the tops of components located under the cab. All openings in the cab floor shall be sealed by boots or mats. Dome light and horn shall be supplied. Cab shall be fabricated in accordance with ANSI B30.5.

3.4.1 Seats. A heavy duty seat, in accordance with SAE J899, shall be provided; material covering shall be black vinyl, black nylon or a combination of both. When material coverings are used in combination the nylon portion shall face operator. The seat shall have a minimum of three inches vertical adjustment. A seat belt shall be provided.

3.4.2 Doors. Doors shall be in accordance with ANSI B30.5.

3.4.3 Windshield and windows. Windshield and windows shall be in accordance with ANSI B30.5. A sun roof shall be provided.

3.4.3.1 Windshield, wipers and washer. One or two front electric windshield wipers and one rear windshield wiper or one skylight wiper shall be provided. Wipers shall be self parking.

3.4.3.2 Windows. Windows shall be operable in order to provide outside air for ventilation.

3.4.4 Mirrors. An adjustable rearview mirror of convex design and not less than 50 square inches shall be mounted on each side of the vehicle on a retractable bracket. The mirrors shall be positioned so that the driver can clearly observe the roadway and vehicles approaching from the rear on either side. A spotter mirror at least five inches in diameter shall be mounted below each rearview mirror.

3.4.5 Instruments and controls. Instruments and controls shall be located within the limits specified in ANSI B30.5, arranged so that those which are most vital to operation have the most predominant location. They shall be placed in blocks distinctly marked to group those instruments and controls pertinent to the operation of any individual component. For example, all instruments and controls relative to the operation of the engine shall be in one block. Controls having similar functions shall have similar actions. Indicator lights shall be green to indicate proper operation and red when warning is intended. All instruments and controls shall be illuminated with nonglare red lighting to be readily visible for night operation. All controls shall be designed so that they are automatically neutralized in the absence of an external force. All controls, including engine clutch, shall meet ANSI B30.5. Instrumentation and electrical controls shall have a quick disconnect plug integrated into the wiring harness to allow removal of the complete panel or sections of the panels for maintenance. At least the following instruments and controls shall be provided:

a. Odometer.
b. Tachometer.
c. Engine oil pressure gauge.
d. Engine coolant temperature gauge.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:47Z
Check the source to verify that this is the current version before use.

e. Transmission oil temperature gauge.
f. Voltmeter or ammeter.
g. Fuel level gauge.
h. Hourmeter.
i. Hydraulic oil temperature gauge.

3.4.6 Climate control system. The climate control system shall provide heater and defroster modes. When required (see 6.2), air conditioning shall be provided. The system shall include a heater core, a blower, a refrigerant compressor, a condenser, an evaporator, and the plumbing ductwork, dampers, controls, and other components necessary to meet the performance requirements specified herein. The climate control system shall induct at least 240 cubic feet per minute of recirculated air into the cab. At least four adjustable panel louvers shall be provided, approximately evenly spaced across the width of the cab; each shall provide approximately equal flow. Cab mounted components shall be protected from inadvertent damage by personnel.

3.4.6.1 Heater and defroster. A cab heater shall be provided with a circulating fan for window defrosting. The heater shall maintain a temperature of at least 40°F at the floor with an outside ambient temperature of -65°F. The defroster shall keep at least 75 percent of the windshield and side windows clear with an outside ambient temperature of -65°F. The defroster shall be in accordance with FMVSS 103.

3.4.6.2 Air conditioner. The air conditioner shall comply with and perform per SAE J1129 except that the minimum amount of recirculated air inducted into the cab shall be 240 cubic feet per minute. Evaporative cooling devices shall not be used. All equipment necessary for replenishment of refrigerant shall be attached to system componentry, only the refrigerant replenishment container shall not be attached to vehicle.

3.5 Electrical system. The vehicle shall be equipped with a 12 volt or 24 volt electrical system; when specified (see 6.2) a 24 volt starting system may be provided. Wiring shall be weatherproof and color coded or numbered for easy identification. All circuits shall be protected by circuit breakers or fuses located in the cab so as to be readily accessible to a mechanic. A master switch shall be provided to disconnect the batteries from the electrical system. A slaving system shall be provided. A receptacle located near the batteries on the exterior of the vehicle to accept either 12 volt or 24 volt, depending on starting system, shall be provided. Slaving system shall charge vehicle batteries and start vehicle. Voltage shall be identified on the receptacles.

3.5.1 Alternator. The engine shall be equipped with an alternator sufficient for operating all vehicle electrical components simultaneously. Minimum capacity of the alternator shall be 130 amperes.

3.5.2 Batteries. The batteries shall supply a minimum of 1,800 cold cranking amp. The battery shall conform to SAE J537J and shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131. When specified (see 6.2), the battery(s) shall be dry charged in accordance with W-B-131, without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture.

3.5.3 Lighting.

3.5.3.1 Vehicle lighting. The vehicle shall be equipped with two sealed beam headlights, taillights, parking lights, clearance lights, brake lights, turn signal lights and reflectors in accordance with Federal Motor Carrier Safety Regulations. The turn signal operating units shall have visible and audible flash indicators; turn signals shall be self-cancelling or manual. Lights and reflectors shall not be mounted on the vertical surface of rub rails or on vehicle bumpers.

3.5.3.2 Interior lighting. The cab and all compartments shall be provided with lighting, either compartmental or exterior, adequate to enable personnel to readily obtain equipment and perform all functions necessary during night operation. Sealed contact switches shall be provided to automatically illuminate the lights in the cab when its door is opened.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.5.3.3 Boom and work area lighting. The cranes shall be equipped with a minimum of two boom lights to illuminate load, one work light on cab to illuminate work area and one spotlight capable of 360° rotation and 180° elevation. All lights shall be mounted in protected locations.

3.5.3.4 Beacon. A pulsating light with amber lens shall be mounted in the center of the cab roof. The control switch shall be mounted on the instrument panel. The beacon light shall be the same as furnished commercially by the contractor.

3.6 Outriggers. Four hydraulically operated outriggers shall be provided, operable separately or in any combination from the operator station. The outriggers shall be capable of extending not less than eight inches below ground level. The outrigger pad size shall be such that the maximum pad force does not exceed 215 pounds per square inch (psi) ground pressure. When specified (see 6.2), a larger outrigger pad may be ordered for muddy terrain use, 40-45 inch diameter. The outriggers shall not collapse due to loss of hydraulic pressure. Outrigger legs and pads shall be sized to support the working loads without loss of crane stability or causing excess ground bearing pressure. An indicator light, located in the cab, shall indicate when outriggers beams are extended. A manual bypass shall be supplied to retract outriggers when total vehicle failure occurs.

3.7 Winterization system. When specified (see 6.2), the vehicle shall be equipped with a winterization system to permit storage and operation in ambient temperatures as low as -65°F. Heaters shall operate on 110/240 volt alternating current and shall be wired through a step up/step down junction block to a single three-pronged (male) weather-proof slave receptacle for receiving external power and grounding the vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c) (7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for storage of the cable shall be provided in the vehicle cab. A warning light located on the dash shall indicate if the winterization kit is energized and shall operate off the system voltage. Heaters shall be furnished as follows:

a. Coolant heater, 1,500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat with an operating range of 140°F to 160°F shall be installed.
b. Immersion type engine oil heater, 300 watt minimum rating, with 140°F to 160°F thermostat shall be installed in the oil pan through a convenient opening.
c. Battery heater shall be provided having a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as -65°F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.
d. A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -65°F to an outlet temperature of +9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system.

3.8 Auxiliary equipment.

3.8.1 Backup alarm. An audible alarm shall be provided to caution personnel in the area when the vehicle is in reverse gear. The alarm shall be in accordance with SAE J994, Type B.

3.8.2 Winch (load hoist). The hoist winch shall be hydraulic motor driven and shall incorporate an automatic brake capable of holding the maximum rated load. The winch shall be equipped with an automatic cable tensioning device, or equivalent, to ensure sufficient pressure maintained on the cable during light/no

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

load reel-in operation. This device shall not reduce cable life by more than 30 percent. Hydraulically controlled power load lowering shall be provided. For both the Size I and Size II cranes, the power down shall have sufficient capacity to hold a load without slipping on a single part line, equal to or greater than the permissible line pull for the winch. The power load lowering shall provide for positive load control. Winch shall conform to SAE J820 and ANSI B30.5.

3.8.3 Tow winch. A front mounted winch shall be provided. Capacity of the winch shall be a minimum of 15,000 pounds line pull on bare drum. Winch line speed shall not be less than 40 feet per minute (FPM) on bare drum. A minimum of 300 feet of wire rope shall be included with the winch. If winch is non-hydraulic type, a safety shear pin shall be installed in the winch drive to preclude overload. Capacity of the shear pin shall be plus 10 percent, minus zero percent. Failure of this shear pin shall cause no damage to or misalignment of winch components. If winch is the hydraulic type it shall be equipped with a preset relief valve which shall prevent overloading of the winch. The relief valve shall empty back into the system.

3.8.4 Drums. All drums shall be designed to ensure proper spooling of the rope and shall incorporate flanges that comply with ANSI B30.5. The drum diameter shall be in accordance with ANSI B30.5. The drum shall be weatherproof.

3.8.5 Wire rope. The load hoist rope shall be not less than 6 X 19 improved plow steel independent wire rope center having a strength factor of not less than 3.5. Minimum rope diameter shall be 0.50 inch for Size I crane and 0.62 inch for the Size II crane. Sufficient rope shall be provided to allow performance of all lifts specified herein. Wire rope forces shall be determined in accordance with SAE J959.

3.8.6 Hook and block. The hook and block shall be of sufficient weight to ensure payout of the rope with no load on the hook. The hook shall be of the swivel type, with a minimum safety factor of five, based on the material minimum ultimate strength. The hook shall be equipped with a throat opening device. The block assembly shall provide for attachment of rope for not less than two part purchase. Hook and block shall be rated at 25 tons for Size I crane and 50 tons for Size II crane. When required (see 6.2), an auxiliary hook and block (rated for manufacturer's maximum lift) shall be provided.

3.8.7 Anti two block system. An anti two block system switch shall be provided to prevent rope breakage and wedging or binding of the hook when it is at the boom end.

3.8.8 Safety devices. Cranes shall be equipped with a minimum of the following indicating and warning devices located in operator station:

a. Electronic load indicator.
b. Electronic angle indicator.
c. Drum rotation indicator.
d. Digital radii indicator.
e. Load moment indicator (LMI) with automatic lockout. This device shall perform the following functions for any configuration of boom length, parts of line, on rubber or outriggers, over side or rear and main or auxiliary hoist:

(1) Provide a dial or digital reading of load moment relative to maximum allowable load moment.
(2) Provide a visual warning when load moment reaches 90 percent of capacity and an audible warning and lockout at 100 percent capacity.

The device shall be powered by the crane's 12 volt electric system. A selector switch with a minimum of two positions shall be provided to set the device for lifting configuration.The load moment device shall conform to SAE J159, J375, and J376.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.8.9 Pintle hook. A nonswivel pintle, conforming to MS51117, shall be mounted on the front and rear of the crane carrier.

3.8.10 Fire extinguisher. A Halon 1211 fire extinguisher (five pounds), Amerex Part Number 355T, or equal, shall be securely installed in a location readily accessible to the operator.

3.8.11 Sheaves. Sheaves shall have a pitch diameter of at least 16 times the rope diameter and shall be of correct size for the wire rope furnished and adequately guarded to prevent rope from leaving the sheave during operation.

3.9 Finish and protective coatings.

3.9.1 Cleaning, painting, plating, anodic films and chemical treatments. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808, except that commercial (see 6.3.1) components may be prepared in accordance with the manufacturers's standard commercial practice, provided it is compatible with the exterior finish paint.

3.9.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defect and shall pass the wet tape test described in 4.6.1.1 of this specification.

3.9.2.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint, Forest Green, Color Number 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or be DuPont Imron, or equal. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be the same color as the exterior finish of the vehicle. Painted materials shall be capable of withstanding a five percent concentrated salt spray (fog) test for 120 hours in accordance with ASTM B 117.

NOTE

"Equal" is defined as having performance identical to or superior to DuPont Imron when subjected to the following tests:

a. Tests in accordance with MIL-C-46168:

(1) Paragraph 4.3.17 Flexibility.

(2) Paragraph 4.3.23 Accelerated weathering.

b. Tests in accordance with MIL-C-83286:

(1) Paragraph 4.8.6 Low temperature resistance.

(2) Paragraph 4.8.10 Resistance to taping.

(3) Paragraph 4.8.12 Pencil hardness.

3.9.2.2 Desert sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be MIL-C-83286, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the vehicle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.9.3 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with paragraph 5.8 of FED-STD-297D. The first article units shall not be rustproofed until after approval of the first article test report.

3.10 Markings. Unless otherwise specified, markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.10.1 Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.11 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 an d be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.11.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.11.2 Control markings. All controls, valves, gauges and indicators used in the operation of the vehicle shall be identified by nameplates securely installed in clearly visible locations.

3.11.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.11.3.1 Loading capacity chart. A permanent chart indicating the safe load capacities for each boom reach and radius for 360° operation with and without outriggers and for pick and carry shall be mounted in clear view of the operator inside the cab. Load capacities shall be based on 85 percent of tipping as determined in accordance with SAE J765. From 20 foot to 35 foot radii, the chart shall be in one foot increments.

3.11.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.11.5 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center of gravity of the vehicle in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

3.12 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.13 Parts control. The contractor shall participate in Government Industry Data Exchange Program (GIDEP) in accordance with MIL-STD-1556b. In the design of the vehicle, the contractor shall ensure the components selected have not been identified as a problem or unresolved deficiency by GIDEP. The contractor shall submit an Alert/Safe-Alert, on a DD Form 1939, whenever an actual or potential problem is evident with respect to a part, component, material, test equipment, process or safety condition. The contractor shall certify maximum use of off-the-shelf items in accordance with MIL-STD-965a.

3.14 Fabrication.

3.14.1 Welders and welding.

3.14.1.1 Welders. All welders employed in the fabrication of the vehicle, except for the commercial chassis, shall be certified in accordance with the applicable requirements of the American Welding Society.

3.14.1.2 Welding. Welded and brazed fabrication shall be in accordance with AWS D14.3, and all welders and welding equipment shall be certified by the requirements of AWS D14.3 in accordance with ANSI B30.5.

3.14.2 Screw assemblies. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.14.3 Riveting. Riveting operations shall be performed to ensure tight fit and complete heading of the rivets.

3.14.4 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable with interference, tight spots, loose spots or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

3.14.5 Cleaning. The vehicle shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts, nuts and washers, rust metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.15 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting; marking of parts and assemblies, alignment of parts and tightness of assembly fasteners, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the vehicle conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. This inspection and testing of the vehicle shall be classified as follows:

a. First article testing (see 4.4)

b. Acceptance testing (see 4.5)

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Transmission oil temperature.

i. Vehicle odometer reading at the start and finish of each test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

j. Engine hourmeter reading at the start and finish of each test.

k. Hydraulic fluid pressure at the pump outlet.

l. Hydraulic fluid temperature in the reservoir.

4.3.3 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubrication or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service or maintenance.

i. Overheating of the engine, transmission or any other vehicle component.

4.4 First article testing.

4.4.1 First article test samples. The first article test sample may be preproduction items or production items which meet the requirements of this specification. The sample shall be identical to the items procured under the terms of the contract. Approval of the first article samples shall not constitute a waiver by the Government of its rights under the provisions of the contract. The first article sample shall be produced and tested in accordance with contractor test sample (see 4.4.2.1).

4.4.2 First article tests.

4.4.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described in 4.6.1 through 4.6.9. All tests shall be run using JP-8 fuel conforming to MIL-T-83133.

4.4.3 Notification of tests. The procuring activity shall be notified of the time and location of all first article tests so that representatives of the procuring activity may attend. The notification shall be in writing and shall include the nomenclature and the national stock number.

4.5 Acceptance testing. Acceptance testing shall be classified by individual testing (see 4.5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

4.5.1 Individual testing. Each vehicle shall be subjected to the tests specified in 4.6.1 and 4.6.6.2.

4.5.1.1 Rejection and retest. When a vehicle fails to meet the requirements of this specification, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor and the component that failed is successfully retested to assure compliance with this specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting and markings. Any painted surface of the exterior of the vehicle may be tested in accordance with section 4.6.1.1 of this specification. For the first article and random sample vehicles, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.1.1 Adhesion (wet) tape test. This method covers a procedure suitable for establishing acceptability of intercoat and surface adhesion of the polyurethane coating system. The test will be accomplished on each production end item, e.g., loader, trailer, vehicle.

a. The test is not applicable to a unit where the surface area or contour of the unit is such that the test would be physically impossible to accomplish, e.g., individual components such as motors or engines.

b. Adhesion tests shall be made on the completed exterior finish on the painted area after drying for a period preferably of at least 48 hours. The test will be accomplished on only one area unless failure occurs, then several areas will be checked to determine that a satisfactory level of adhesion has been achieved. Three consecutive failures are grounds for rejection. Failures are defined as any visible peeling, chipping or defects.

c. Perform the wet tape test as follows:

(1) Soak a clean cloth in distilled water (size at least 6 inches X 3 inches).

(2) Fold the cloth into a square while it is still wet and place it on the area to be tested.

(3) Place a piece of plastic over the wet cloth. The plastic must completely cover the cloth and hold the cloth in contact with the test surface.

(4) Tape the plastic sheet onto the structure. Completely seal the edges with tape so that the water cannot evaporate.

(5) Allow the wet cloth to continuously contact the area for a minimum of 24 hours.

(6) Remove the wet cloth and blot up the surface water. Immediately apply a one inch strip of tape, (Code No. 250 Minnesota Mining and Manufacturing Company, age of tape not to exceed one year), adhesive side down. Press the tape down, using two passes of 4.5 pound rubber covered roller or employ firm pressure with the hand.

(7) Remove the tape in one abrupt motion and examine the tested area for any paint damage such as removal of paint at one of the layers of the finish system or removal of the entire system from metal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

4.6.2 Preliminary operation. The following tests shall be performed on the crane:

a. Rotate boom.

b. Extend outriggers.

c. Extend boom to maximum length.

4.6.3 Mobility tests.

4.6.3.1 Maximum speed. Operate crane on level smooth concrete or paved surface without a boom load. Drive the crane the distance necessary to attain maximum speed prior to entering a measured distance. Record time to traverse this distance, distance (feet) and speed (MPH). The crane shall be driven over the measured course at least three times in each direction. The speed shall be the average of the three runs in each direction. Inability of the crane to attain the minimum speed specified in section 3.2 of this specification, failure of any components, overheating, malfunction or leakage of fluid shall be cause for rejection. A fifth wheel is required for this test.

4.6.3.2 Grade operation. A fully loaded crane (maximum load at 20 foot radius) shall be driven up and down a 10 percent dry, paved grade. The maximum speed that can be maintained while climbing the grade shall be observed and shall not be less than 2.0 MPH. Engine oil pressure shall not be less than the manufacturer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the grade.

4.6.3.3 Side slope operation. The fully loaded crane (maximum load at 20 foot radius) shall be driven in both directions on a side slope of 10 percent. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the slope. A maximum lift at 30 foot radius, with outriggers, shall be performed on a 6 percent slope and then rotated 360°.

4.6.3.4 Suspension articulation. The vehicle shall be tested to demonstrate compliance with 3.2.1(k). The sides of the blocks may be sloped; however, the block length shall be such that when one tire on a side is at the top of the block, the other tire on that side is on the ground.

4.6.3.5 Brake tests.

4.6.3.5.1 Service brake tests. The vehicle shall be tested in accordance with SAE J1152 to demonstrate compliance with 3.3.3.1.

4.6.3.5.2 Parking brake tests. The vehicle shall be tested in accordance with SAE J1152 to demonstrate compliance with 3.3.3.2.

4.6.4 Performance tests.

4.6.4.1 Lifting capacity tests. Lifting capacities of the crane shall be determined by subjecting the crane to load stability tests in accordance with SAE J765. The tests shall be conducted at the minimum working radius and at three intermediate, equidistant radii, with and without outriggers, when the boom is retracted to its minimum length. Conformance to 3.2.2 shall be determined by multiplying the balance point load by 85 percent for outriggers and 75 percent for on rubber. Where the lifting capacity is based on structural strength or available hoisting power rather than resistance to overturning, the manufacturer shall submit his test procedure for approval. The procedures shall include all applicable provisions of SAE J765 and shall be designed to show that the crane will lift at least 115 percent with outriggers and at least 125 percent without outriggers of the rated capacity loads specified (see 3.2.2). Measurements at five radii shall be made and shall agree with the boom angle chart (see 3.9.1.4) for both outriggers and on rubber lifts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

4.6.4.2 Rotating load interference test. Ability of crane to lift aircraft, called out in paragraph 3.2.2, and rotate at 8 foot load elevation through a 360° rotation shall be demonstrated analytically. The analytical method shall consist of the proper drawings, charts, diagrams, and calculations necessary to validate the cranes compliance.

4.6.4.3 Power load lowering test. General inspection of the power down system shall be made for adequate lubrication, ease of operation of controls, safety precautions and for good manufacturing practices. The power load lowering system shall be tested as follows to determine conformance to 3.8.2.

a. With the crane equipped with a boom of a length as specified in 3.1.9 and with outriggers set, position the revolving superstructure in the most stable craning position and assemble a single line load equal to the maximum load divided by the number of parts of hoist line reeving.

b. Position the boom angle for the minimum working radius.

c. Hoist the load 20 feet above the ground level, plus three inches, or until interference with the crane boom is obvious.

d. Prepare the crane for power down operation on the load line.

e. Operate the power load lowering system in accordance with manufacturer's instructions and lower the load under power. Vary the speed of the load.

f. When the load reaches the ground, immediately repeat operations c, d and e. These operations shall be repeated until 20 cycles have been completed.

g. Repeat the above tests using the jib and auxiliary hoist.

4.6.5 Design verification test.

4.6.5.1 Backward stability test. The crane shall be tested for weight distribution to insure proper backward stability while equipped with the shortest recommended boom. Set a minimum radius with hook, hook block and load handling equipment resting on the ground and with outriggers retracted and free of the ground. Using platform scales accurate to within plus or minus one percent of the observed vehicle weight, with surrounding terrain level and of the same elevation as top of scales platform, measure the following:

a. Total weight. Check to see that the working weight is conspicuously stenciled on each side of the crane.

b. Weight on the wheels on the right side with boom at right angles to the carrier over the right side and at minimum radius. Determine that this weight is not less than the specified minimum (see 3.1.5).

c. Weight on the wheels on the left side with the boom at right angles to the carrier over the left side and at minimum radius. Determine that this weight is not less than the specified minimum (see 3.1.5).

d. Weight on the front wheels with the boom on the carrier axis over the front wheels and at minimum radius. Determine that this weight is not less than the specified minimum (see 3.1.5).

4.6.5.3 Electromagnetic interference measurement. The vehicle and all electrical components shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.1.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

4.6.5.4 Sound level testing. The vehicle shall be tested in accordance with SAE J919a to demonstrate compliance with 3.1.8.

4.6.5.5 Emergency steering test. The emergency steering system specified in 3.3.2 must be run, tested and conform to SAE J53.

4.6.5.6 Crane hook test. The crane hook shall be given a wet or dry ferromagnetic particle test to determine surface or subsurface cracks or discontinuities. Either method of test used shall show clearly visible particles defining existing discontinuities. Any discontinuities shall be cause for rejection.

4.6.5.7 Safety devices test. The safety devices shall be tested to determine conformance to 3.8.8.

4.6.5.8 Corrosion test. A three foot by two foot panel, rustproofed in accordance with the contractor's standard commercial practice and painted in accordance with 3.9.2.1 of this specification, shall withstand 120 hours of a five percent concentrated salt spray in a closed environment without any evidence of corrosion.

4.6.5.9 Swing brake test. Test the swing brake at an engine speed of 600 RPM by engaging the brake with the superstructure in several positions and load weights and attempt to rotate the superstructure. A crane equipped with swing brakes shall have the mechanism engaged to determine conformance to these requirements. Check to insure that the superstructure will remain stationary when the boom is positioned in the boom rack.

4.6.5.10 Winch performance. The crane shall be tested in accordance with SAE J820 to demonstrate compliance with 3.8.2.

4.6.5.11 Crane structural test. The crane shall be tested in accordance with SAE J1063. If the crane is a commercial item which has previously satisfied this requirement, a copy of the previous test report, including all data sheets, may be provided in lieu of retesting.

4.6.6 Environmental tests.

4.6.6.1 Low temperature storage and operation. The complete vehicle including all accessories, shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.4(a). Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with paragraph 4.6.6.2. The vehicle shall be cold soaked for a period of not less than 24 hours in an ambient temperature of -65°F, plus or minus 2.5°F. Ether injection, glow plugs as well as specific heaters stated in 3.3.1.1.2, shall be the only starting aids used. Arctic fuel and lubricants shall be permitted at this temperature. At this time, the vehicle shall be tested for operation of all lights, brakes, gears and other controls to prove proper working condition. The vehicle shall then be cold soaked for a period of 24 hours at 0°F, plus or minus 2.5°F. Ether injection shall be the only starting aid used. Noncompliance with said test shall be cause for rejection. A fine mist of water shall be applied to the entire windshield to form a thin film of ice. The defroster shall clear the windshield within one hour and maintain clear vision throughout the remainder of the test. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site.

4.6.6.2 Operational checkout. All controls, levers, mechanisms, and doors shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation.

4.6.6.3 Rain. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 506.2, Procedure III, of MIL-STD-810 to demonstrate compliance with 3.1.4(c). Operational checkouts shall be in accordance with 4.6.8.5. The interior of the cab and all compartments shall be examined for water; any evidence of leakage shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

4.6.6.4 Other environmental tests. The components and materials used in the construction of the vehicle shall be subjected to, or certified as having passed, or being capable of passing, the following test methods of MIL-STD-810 to demonstrate compliance with 3.1.4:

a. Humidity - Method 507.2, Procedure I.

b. Salt Fog - Method 509.2, Procedure I. Cycling test shall be used. One cycle consists of 24 hours exposure to the test environment followed by 24 hour exposure to standard ambient conditions. A minimum of 7 cycles shall be conducted.

c. Sand and Dust - Method 510.2, Procedures I and II. For Procedure I, air velocity shall be 1,750 ft/min and high temperature adjustment shall be 125°F. For Procedure II, air velocity shall be 5,700 ft/min; sand concentration of 0.0623 g/ft^3, plus or minus 0.015 g/ft^3; 90 minutes per face duration.

4.6.7 Reliability demonstration. Compliance with the quantitative reliability requirement of 3.1.3.1 shall be demonstrated by the tests in 4.6.7.2 and 4.6.7.3. The vehicle shall complete, as a minimum, the tests described in 4.6.7.2 and 4.6.7.3 with the contractor's option of using the tests described in paragraphs 4.6.3, 4.6.4, 4.6.5.5, 4.6.5.9, 4.6.5.10, and 4.6.5.11 towards the total hours required to meet the reliability demonstration. The required 250 hours MTBF at a 0.80 confidence level equates with 400 hours of testing without a failure. The tests in 4.6.7.2 and 4.6.7.3 shall be continued if necessary to accumulate the required hours. Scheduled maintenance, maintenance events, or adjustment authorized by this specification or the procuring activity shall not be counted as failures.

4.6.7.1 Vehicle durability test.

4.6.7.2 Endurance test. The crane shall lift the maximum rated pick and carry load at a 30 foot radius. The load shall be carried over several surfaces for a distance of not less than fifteen miles. The average speed for this test shall be 1.5 MPH minimum and the distance on each surfaces are as follows:

SPEED (MPH)	SMOOTH PAVEMENT	ROUGH TERRAIN
1.5 MPH	6 MILES	9 MILES (A MINIMUM OF 35 VEHICLE CONE INDEX)

The change of surfaces shall occur without stopping. At the conclusion of the durability test, the crane shall be inspected and any damages recorded.

4.6.7.3 Cross country operation. The unloaded crane shall be driven over cross country terrain for at least 150 miles at an average speed of not less than 10 MPH to demonstrate compliance with 3.2.1.

4.6.8 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.1.3.2 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent nor more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation or miles traveled. The capability of performing maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

4.7 Refurbishment. After successful completion of all required testing, each vehicle shall be restored to like new condition.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the instructions of the procuring activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cranes furnished under this document will be used for crashed aircraft recovery, Size I, F-15, Size II, F111. Three feet clearance and no outrigger interference shall be maintained during all lifts.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, crane size, and date of this specification.

b. Finish color required, if not Forest Green (see 3.9.2.1).

c. When spare wheel and tire are required (see 3.3.6).

d. When auxiliary hook and block is required (see 3.8.6).

e. Identification markings required (see 3.10.1).

f. When dry charged batteries are required (see 3.5.2).

g. When a 24 volt starting system is required (see 3.5).

h. When air conditioning is required (see 3.4.6.2).

i. Winterization, if required (see 3.7).

j. Ozone resistant tires, if required (see 3.3.6).

k. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

6.2.1 First article. When a first article inspection is required, the items should be first article samples. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.2.1.1 Test report. The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedure, as appropriate. Test conditions at the time of failures or irregular operation identified shall be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.

g. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

h. Test report for each component covered by a document listed in the DODISS.

i. Plan for testing production units.

6.3 Definitions.

6.3.1 Commercial. A manufacturer's standard (product), having been marketed, in production, and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototype, or experimental units, do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

6.3.2 Cross country terrain. Cross country terrain consists of open fields, broken ground and uneven terrain in general.

6.4 Data requirements. The following Data Item Descriptions (DID's) must be listed, as applicable, on the contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
	DI-A-7088	Conference Agenda	See DD Form 1423
	DI-A-7089	Conference Minutes	See DD Form 1423
	DI-A-4901	Test Procedures	See DD Form 1423
6.2.1.1	DI-A-4902	Test Report	See DD Form 1423
4.4.3	DI-T-2373A/T	Notification of Tests	See DD Form 1423
	DI-A-3011	Still Photo Coverage	See DD Form 1423

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-1, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing.

Block
Boom
Cross country
Drums
Hook
Outriggers
Rope, wire
Sheaves
Winch

Custodian:
Air Force - 99

Preparing activity:
Air Force - 84

(Project 3810-F102)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-83821 (84)	2. DOCUMENT DATE (YYMMDD) 90-11-26

3. DOCUMENT TITLE
CRANE, WHEEL MOUNTED, CRASH, ROUGH TERRAIN, GYDRAULIC FULL REVOLVING, 4X4, DED-

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	912-926-5616	468-5616
c. ADDRESS (Include Zip Code) WR-ALC/MMDETB Robins AFB GA 31098-5609	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-0-26T04:47Z
Check the source to verify that this document is authentic before use.